

Операционные системы

Основы интерфейса взаимодействия пользователя с системой Unix на уровне командной строки

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1. Цели и задачи работы
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1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.

1.2 Задачи лабораторной работы

- 1 Определить имя и путь домашнего каталога
- 2 Изучить команду `ls`.
- 3 Выполнить действия с каталогами.
- 4 Получить дополнительные сведения при помощи справки по командам.
- 5 Изучить команду `history`.

2. 2. Процесс выполнения лабораторной работы

2.1 Имя и путь к домашнему каталогу

```
iaatabaeva@iaatabaeva:~$ cd  
iaatabaeva@iaatabaeva:~$ pwd  
/home/iaatabaeva  
iaatabaeva@iaatabaeva:~$
```

Рисунок 1: Путь к домашнему каталогу

2.2 Опции команды ls

```
iaatabaeva@iaatabaeva:~$ cd /tmp
iaatabaeva@iaatabaeva:/tmp$ ls
c39fcee1e-e876-4363-aa05-a03d8a26e272.zip
e9b150a8-c8a5-4a04-a153-36617035c568.zip
happd.sock
snap-private-tmp
systemd-private-a0cad7ae4f9947649a06d0d78363e384-abrt.service-z48S9n
systemd-private-a0cad7ae4f9947649a06d0d78363e384-chrond.service-I53i7I
systemd-private-a0cad7ae4f9947649a06d0d78363e384-colord.service-KHEr3t
systemd-private-a0cad7ae4f9947649a06d0d78363e384-dbus-broker.service-dPkJQX
systemd-private-a0cad7ae4f9947649a06d0d78363e384-fwupd.service-MWF0sY
systemd-private-a0cad7ae4f9947649a06d0d78363e384-irqbalance.service-Zv28fU
systemd-private-a0cad7ae4f9947649a06d0d78363e384-low-memory-monitor.service-M3NX8p
systemd-private-a0cad7ae4f9947649a06d0d78363e384-ModemManager.service-89XLBF
systemd-private-a0cad7ae4f9947649a06d0d78363e384-pcscd.service-0laNf4
systemd-private-a0cad7ae4f9947649a06d0d78363e384-polkit.service-JcbayQ
systemd-private-a0cad7ae4f9947649a06d0d78363e384-rtkit-daemon.service-suoiDP
systemd-private-a0cad7ae4f9947649a06d0d78363e384-switcheroo-control.service-H4a4Ck
systemd-private-a0cad7ae4f9947649a06d0d78363e384-systemd-logind.service-NXNy1K
systemd-private-a0cad7ae4f9947649a06d0d78363e384-systemd-resolved.service-p0u6o9
systemd-private-a0cad7ae4f9947649a06d0d78363e384-upower.service-lSmB3j
'tcU0WK5PwneH++Y9c0o680J7cW+IVVsFTUXq6YhU0V8='
tmp-bc.xpi
vmware-root_894-2730693566
iaatabaeva@iaatabaeva:/tmp$
```

Рисунок 2: Команда ls

2.3 Опции команды ls

```
iaatabaeva@iaatabaeva:/tmp$  
iaatabaeva@iaatabaeva:/tmp$ ls -la  
.  
..  
c39fce1e-e876-4363-aa05-a03d8a26e272.zip  
e9b150a8-c8a5-4a04-a153-36617035c568.zip  
.font-unix  
happd.sock  
.ICE-unix  
snap-private-tmp  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-abrt.service-z48S9n  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-chronyd.service-I53i7I  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-colord.service-KHER3t  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-dbus-troker.service-dPkJQX  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-fwupd.service-MWFOsY  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-irqbalance.service-Zv28fU  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-low-memory-monitor.service-M3NX8p  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-ModemManager.service-89XLBF  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-pcsd.service-0laNf4  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-polkit.service-JcbayQ  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-rtkit-daemon.service-suoiDP  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-switcheroo-control.service-H4a4Ck  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-systemd-logind.service-NXNy1K  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-systemd-resolved.service-p0u6o9  
systemd-private-a0cad7ae4f9947649a06d0d78363e384-upower.service-lSmB3j  
'tcU0WK5PwneH++Y9c0o680J7cW+IVVsFTUXq6YhUOV8='  
tmp-bc.xpi  
vmware-root_894-2730693566  
.X0-lock  
.X11-unix
```

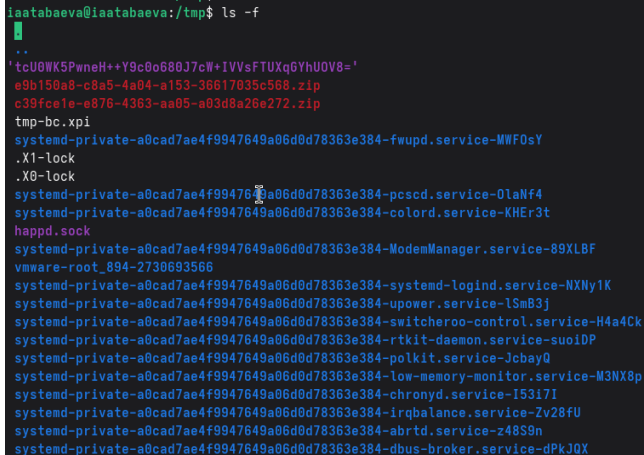
Рисунок 3: Команда ls -la

2.4 Опции команды ls

```
iaatabaeva@iaatabaeva: /tmp$  
iaatabaeva@iaatabaeva: /tmp$ ls -l  
total 9672  
-rw-r--r--. 1 iaatabaeva iaatabaeva 4182728 Mar 6 13:07 c39fce1e-e876-4363-aa05-a03d8a26e272.zip  
-rw-r--r--. 1 iaatabaeva iaatabaeva 4182728 Mar 6 13:07 e9b150a8-c8a5-4a04-a153-36617035c568.zip  
srw-rw-rw-. 1 root      root              0 Mar 6 13:06 happd.sock  
drwx-----. 2 root      root              40 Mar 6 13:05 snap-private-tmp  
drwx-----. 3 root      root              60 Mar 6 13:05 systemd-private-a0cad7ae4f9947649a06d0d7  
8363e384-abrt.d.service-z48S9n  
drwx-----. 3 root      root              60 Mar 6 13:05 systemd-private-a0cad7ae4f9947649a06d0d7  
8363e384-chronyd.service-I53i7I  
drwx-----. 3 root      root              60 Mar 6 13:06 systemd-private-a0cad7ae4f9947649a06d0d7  
8363e384-colord.service-KHEr3t  
drwx-----. 3 root      root              60 Mar 6 13:05 systemd-private-a0cad7ae4f9947649a06d0d7  
8363e384-dbus-broker.service-dPkJQX  
drwx-----. 3 root      root              60 Mar 6 13:06 systemd-private-a0cad7ae4f9947649a06d0d7  
8363e384-fwupd.service-MWF0sY  
drwx-----. 3 root      root              60 Mar 6 13:05 systemd-private-a0cad7ae4f9947649a06d0d7  
8363e384-irqbalance.service-Zv28fU  
drwx-----. 3 root      root              60 Mar 6 13:05 systemd-private-a0cad7ae4f9947649a06d0d7  
8363e384-low-memory-monitor.service-M3NX8p  
drwx-----. 3 root      root              60 Mar 6 13:06 systemd-private-a0cad7ae4f9947649a06d0d7  
8363e384-ModemManager.service-89XLBF  
drwx-----. 3 root      root              60 Mar 6 13:06 systemd-private-a0cad7ae4f9947649a06d0d7  
8363e384-openssh.service-01aHf4
```

Рисунок 4: Команда ls -l

2.5 Опции команды ls



```
iaatabaeva@iaatabaeva:/tmp$ ls -f
..
'tcU0WK5PwneH++Y9c0o680J7cW+IVVsFTUXq6YhUOV8='
e9b150a8-c8a5-4a04-a153-36617035c568.zip
c39fce1e-e876-4363-aa05-a03d8a26e272.zip
tmp-bc.xpi
systemd-private-a0cad7ae4f9947649a06d0d78363e384-fwupd.service-MWF0sY
.X1-lock
.X0-lock
systemd-private-a0cad7ae4f9947649a06d0d78363e384-pcscd.service-0laNf4
systemd-private-a0cad7ae4f9947649a06d0d78363e384-colord.service-KHEr3t
happd.sock
systemd-private-a0cad7ae4f9947649a06d0d78363e384-ModemManager.service-89XLBF
vmware-root_894-2730693566
systemd-private-a0cad7ae4f9947649a06d0d78363e384-systemd-logind.service-NXNy1K
systemd-private-a0cad7ae4f9947649a06d0d78363e384-upower.service-lSmB3j
systemd-private-a0cad7ae4f9947649a06d0d78363e384-switcheroo-control.service-H4a4Ck
systemd-private-a0cad7ae4f9947649a06d0d78363e384-rtkit-daemon.service-suoiDP
systemd-private-a0cad7ae4f9947649a06d0d78363e384-polkit.service-JcbayQ
systemd-private-a0cad7ae4f9947649a06d0d78363e384-low-memory-monitor.service-M3NX8p
systemd-private-a0cad7ae4f9947649a06d0d78363e384-chronyd.service-I53i7I
systemd-private-a0cad7ae4f9947649a06d0d78363e384-irqbalance.service-Zv28fU
systemd-private-a0cad7ae4f9947649a06d0d78363e384-abrttd.service-z48S9n
systemd-private-a0cad7ae4f9947649a06d0d78363e384-dbus-broker.service-dPkJQX
```

Рисунок 5: Команда ls -f

2.6 Каталог /var/spool

```
iaatabaeva@iaatabaeva:/tmp$ cd /var/spool/  
iaatabaeva@iaatabaeva:/var/spool$ ls -l  
total 0  
drwxr-x--x. 1 root abrt 1522 Mar  6 13:09 abrt  
drwx-----. 1 abrt abrt   0 Dec  4 03:00 abrt-upload  
drwxr-xr-x. 1 root root  66 Oct 23 07:05 anacron  
drwx-----. 1 root root  18 Oct 23 07:07 at  
drwx-----. 1 root root   0 Aug  6 2025 cron  
drwx--x---. 1 root lp    6 Dec 12 03:00 cups  
drwxr-xr-x. 1 root root   0 Jul 30 2025 lpd  
drwxrwxr-x. 1 root mail 1358 Mar  5 15:40 mail  
drwxr-xr-x. 1 root root   0 Jan 27 03:00 plymouth  
iaatabaeva@iaatabaeva:/var/spool$
```

Рисунок 6: Каталог /var/spool

2.7 Домашний каталог

```
iaatabaeva@iaatabaeva:~/var/spool$  
iaatabaeva@iaatabaeva:/var/spool$ cd  
iaatabaeva@iaatabaeva:~$ ls  
Desktop    Downloads  Music      Public    Templates work  
Documents  git-extended Pictures   site      Videos  
iaatabaeva@iaatabaeva:~$ ls -al  
total 24  
drwx-----. 1 iaatabaeva iaatabaeva 374 Feb 16 13:43 .  
drwxr-xr-x. 1 root      root      1358 Mar  5 15:40 ..  
-rw-----. 1 iaatabaeva iaatabaeva 5722 Feb 16 14:23 .bash_history  
-rw-r--r--. 1 iaatabaeva iaatabaeva  18 Jul 23  2025 .bash_logout  
-rw-r--r--. 1 iaatabaeva iaatabaeva  144 Jul 23  2025 .bash_profile  
-rw-r--r--. 1 iaatabaeva iaatabaeva  680 Feb 16 13:28 .bashrc  
drwx-----. 1 iaatabaeva iaatabaeva  512 Mar  6 13:10 .cache  
drwx-----. 1 iaatabaeva iaatabaeva  368 Mar  6 13:11 .config  
drwxr-xr-x. 1 iaatabaeva iaatabaeva   0 Feb 16 13:16 Desktop  
drwxr-xr-x. 1 iaatabaeva iaatabaeva   0 Feb 16 13:16 Documents  
drwxr-xr-x. 1 iaatabaeva iaatabaeva  26 Mar  6 13:11 Downloads  
-rw-r--r--. 1 iaatabaeva iaatabaeva 232 Feb 16 13:22 .gitconfig  
drwxr-xr-x. 1 iaatabaeva iaatabaeva  74 Feb 16 13:33 git-extended  
drwx-----. 1 iaatabaeva iaatabaeva  136 Feb 16 13:21 .gnupg  
drwx-----. 1 iaatabaeva iaatabaeva  20 Feb 16 13:16 .local  
drwxr-xr-x. 1 iaatabaeva iaatabaeva  34 Oct 23 07:05 .mozilla  
drwxr-xr-x. 1 iaatabaeva iaatabaeva   0 Feb 16 13:16 Music  
drwxr-xr-x. 1 iaatabaeva iaatabaeva  26 Feb 16 13:43 .npm  
drwxr-xr-x. 1 iaatabaeva iaatabaeva   0 Feb 16 13:16 Pictures  
drwxr-xr-x. 1 iaatabaeva iaatabaeva   0 Feb 16 13:16 Public
```

Рисунок 7: Файлы в домашнем каталоге

2.8 Работа с каталогами

```
iaatabaeva@iaatabaeva:~$  
iaatabaeva@iaatabaeva:~$ mkdir newdir  
iaatabaeva@iaatabaeva:~$ mkdir newdir/morefun  
iaatabaeva@iaatabaeva:~$ mkdir letters memos misk  
iaatabaeva@iaatabaeva:~$ ls  
Desktop  Downloads  letters  misk  newdir  Public  Templates  work  
Documents  git-extended  memos  Music  Pictures  site  Videos  
iaatabaeva@iaatabaeva:~$ rm letters/ memos/ misk/  
rm: cannot remove 'letters/': Is a directory  
rm: cannot remove 'memos/': Is a directory  
rm: cannot remove 'misk/': Is a directory  
iaatabaeva@iaatabaeva:~$ rm -r letters/ memos/ misk/  
iaatabaeva@iaatabaeva:~$  
iaatabaeva@iaatabaeva:~$ rm -r newdir/  
iaatabaeva@iaatabaeva:~$ ls  
Desktop  Downloads  Music  Public  Templates  work  
Documents  git-extended  Pictures  site  Videos  
iaatabaeva@iaatabaeva:~$
```

Рисунок 8: Действия с каталогами

2.9 Опции команды ls

```
./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources':
csl tex

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/csl':
gost-r-7-0-5-2008-numeric.csl

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/tex':
preamble.tex

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/scripts':
image-report mpv-shot
iaatabaeva@iaatabaeva:~$ ls -t
Downloads  git-extended  Desktop      Music        Public       Videos
site       work          Documents    Pictures     Templates
iaatabaeva@iaatabaeva:~$
```

Рисунок 9: Команда ls -R и ls -t

2.10 Справка по командам

```
BASH_BUILTINS(1)                                General Commands Manual                                BASH_BUILTINS(1)

NAME
: , . , [ , alias, bg, bind, break, builtin, caller, cd, command, compgen, complete, compopt,
continue, declare, dirs, disown, echo, enable, eval, exec, exit, export, false, fc, fg,
getopts, hash, help, history, jobs, kill, let, local, logout, mapfile, popd, printf,
pushd, pwd, read, readarray, readonly, return, set, shift, shopt, source, suspend, test,
times, trap, true, type, typeset, ulimit, umask, unalias, unset, wait - bash built-in com-
mands, see bash(1)

BASH BUILTIN COMMANDS
Unless otherwise noted, each builtin command documented in this section as accepting op-
tions preceded by - accepts -- to signify the end of the options. The :, true, false, and
test/[ builtins do not accept options and do not treat -- specially. The exit, logout,
return, break, continue, let, and shift builtins accept and process arguments beginning
with - without requiring --. Other builtins that accept arguments but are not specified
as accepting options interpret arguments beginning with - as invalid options and require
-- to prevent this interpretation.

: [arguments]
    No effect; the command does nothing beyond expanding arguments and performing any
    specified redirections. The return status is zero.

. [-p path] filename [arguments]
source [-p path] filename [arguments]
    The . command (source) reads and execute commands from filename in the current
    shell environment and returns the exit status of the last command executed from
    filename.

Manual page cd(1) line 1 (press h for help or q to quit)
```

Рисунок 10: Справка по команде cd

2.11 Справка по командам

```
PWD(1)                                User Commands                                PWD(1)

NAME
    pwd - print name of current/working directory

SYNOPSIS
    pwd [OPTION]...

DESCRIPTION
    Print the full filename of the current working directory.

    -L, --logical
        use PWD from environment, even if it contains symlinks

    -P, --physical
        resolve all symlinks

    --help display this help and exit

    --version
        output version information and exit

    If no option is specified, -P is assumed.

    Your shell may have its own version of pwd, which usually supersedes the version described
    here. Please refer to your shell's documentation for details about the options it sup-
    ports.

Manual page pwd(1) line 1 (press h for help or q to quit)
```

Рисунок 11: Справка по команде pwd

2.12 Справка по командам

```
MKDIR(1)                                User Commands                                MKDIR(1)

NAME
    mkdir - make directories

SYNOPSIS
    mkdir [OPTION]... DIRECTORY...

DESCRIPTION
    Create the DIRECTORY(ies), if they do not already exist.

    Mandatory arguments to long options are mandatory for short options too.

    -m, --mode=MODE
        set file mode (as in chmod), not a=rwx - umask

    -p, --parents
        no error if existing, make parent directories as needed, with their file modes un-
        affected by any -m option

    -v, --verbose
        print a message for each created directory

    -Z
        set SELinux security context of each created directory to the default type

    --context[=CTX]
        like -Z, or if CTX is specified then set the SELinux or SMACK security context to
        CTX

Manual page mkdir(1) line 1 (press h for help or q to quit)
```

Рисунок 12: Справка по команде mkdir

2.13 Справка по командам

```
RMDIR(1)                                User Commands                                RMDIR(1)

NAME
    rmdir - remove empty directories

SYNOPSIS
    rmdir [OPTION]... DIRECTORY...

DESCRIPTION
    Remove the DIRECTORY(ies), if they are empty.

    --ignore-fail-on-non-empty
        ignore each failure to remove a non-empty directory

    -p, --parents
        remove DIRECTORY and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b
        a'

    -v, --verbose
        output a diagnostic for every directory processed

    --help display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie.

Manual page rmdir(1) line 1 (press h for help or q to quit)
```

Рисунок 13: Справка по команде rmdir

2.14 Справка по командам

```
RM(1)                                User Commands                                RM(1)

NAME
  rm - remove files or directories

SYNOPSIS
  rm [OPTION]... [FILE]...

DESCRIPTION
  This manual page documents the GNU version of rm.  rm removes each specified file.  By default, it does not remove directories.

  If the -i or --interactive=once option is given, and there are more than three files or the -r, -R, or --recursive are given, then rm prompts the user for whether to proceed with the entire operation.  If the response is not affirmative, the entire command is aborted.

  Otherwise, if a file is unwritable, standard input is a terminal, and the -f or --force option is not given, or the -i or --interactive=always option is given, rm prompts the user for whether to remove the file.  If the response is not affirmative, the file is skipped.

OPTIONS
  Remove (unlink) the FILE(s).

  -f, --force
    ignore nonexistent files and arguments, never prompt

  -i      prompt before every removal

Manual page rm(1) line 1 (press h for help or q to quit)
```

Рисунок 14: Справка по команде rm

2.15 История командной строки

```
276 cd /tmp
277 ls
278 ls -a
279 ls -l
280 ls -f
281 cd /var/spool/
282 ls -l
283 cd
284 ls
285 ls -al
286 mkdir newdir
287 mkdir newdir/morefun
288 mkdir letters memos misk
289 ls
290 rm letters/ memos/ misk/
291 rm -r letters/ memos/ misk/
292 rm -r newdir/
293 ls
294 ls -R
295 ls -t
296 help cd
297 man pwd
298 man mkdir
```

Рисунок 15: Команда history

3. 3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия пользователя с системой посредством командной строки.